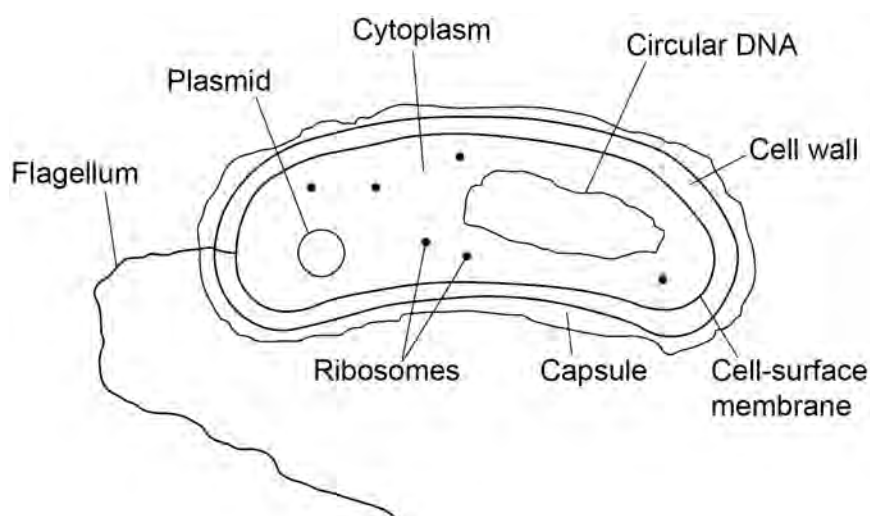


07.1

Figure 11 shows the structure of the bacterium *Vibrio cholerae*.

Figure 11



Name **three** structures shown in **Figure 11** that are **not** found in all bacteria.

[2 marks]

- 1 _____
- 2 _____
- 3 _____

07.2

V. cholerae produces a toxin that only binds to the cell-surface membrane of epithelial cells lining the ileum. The toxin is a protein.

Explain why the toxin can only bind to the cell-surface membrane of epithelial cells in the ileum.

[2 marks]



0	7	.	3
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The toxin causes chloride ion channels in the cell-surface membrane to open. Chloride ions diffuse into the ileum. This causes diarrhoea (watery faeces).

Use your knowledge of water potential to explain how an increase of chloride ions in the ileum results in diarrhoea.

[2 marks]

Question 7 continues on the next page

Turn over ►



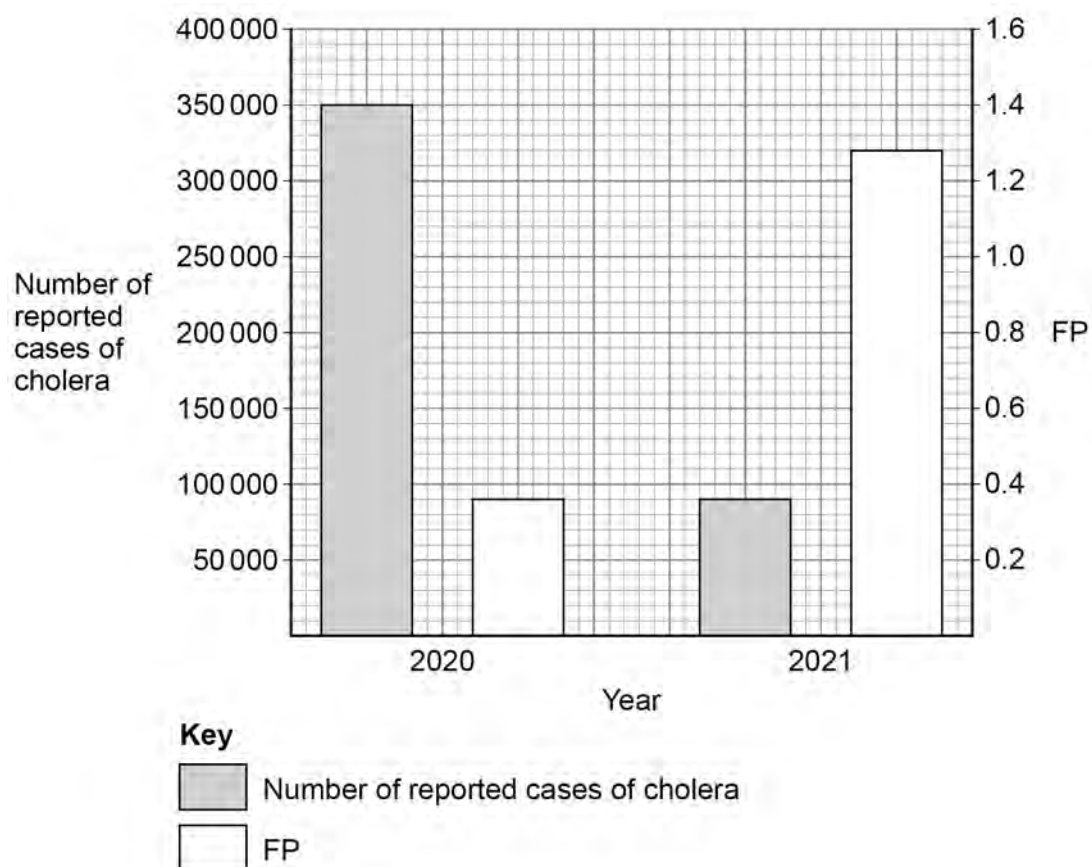
V. cholerae causes the disease cholera.

The Fatality Percentage (FP) is the percentage of people diagnosed with cholera who die from it.

Figure 12 shows:

- the number of reported cases of cholera in a population for the years 2020 and 2021
- the FP for cholera for the years 2020 and 2021.

Figure 12



Do not write
outside the
box



0 7 . 4

Using **Figure 12** and the information provided, calculate the difference in the number of deaths from cholera in 2020 and 2021.

[2 marks]

Difference in the number of deaths _____

0 7 . 5

The calculated numbers of deaths from cholera, in 2020 and 2021, are lower than the true number of deaths.

Using the information suggest **one** reason why.

[1 mark]

9

Turn over for the next question

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